02. Arc Spectra: Atomic Emission Spectra of Copper and Brass

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Roll Number: IMS24267

Date of the experiment: 11 September 2026
Date of Submission: October 11, 2026

1 Name of the Experiment

Arc Spectra: Atomic Emission Spectroscopy of Copper and Brass using a CCD-Based Constant Deviation Spectrometer and Hartmann Dispersion Calibration.

2 Aim of the Experiment

1. To calibrate a prism spectrometer equipped with a linear CCD sensor array using the characteristic emission lines of a low-pressure Mercury (Hg) discharge lamp.
2. To calculate the empirical calibration constants $(A, B, \lambda_0, d_0, C)$ of the Hartmann dispersion relation.
3. To record and analyze the atomic emission arc spectrum produced by pure Copper (Cu) electrodes.
4. To record and analyze the atomic emission arc spectrum produced by Brass electrodes.
5. To determine the wavelengths of prominent spectral lines for both samples using the calibrated Hartmann formula, compare them against standard NIST atomic emission values, and identify the elemental constituents of the Brass specimen.

3 Theory

3.1 Origin of Atomic Emission Spectra

Atomic emission spectroscopy is based on the principle that atoms in an excited state release electromagnetic radiation when their electrons transition from higher energy levels to lower energy levels. According to quantum mechanics, the energy levels of electrons bound to an isolated atomic nucleus are discrete and quantised. When an electron drops from an initial excited state of energy E_2 to a lower state of energy E_1, a photon is

emitted. The frequency ν and wavelength λ of the emitted photon are governed by the Planck-Einstein relation:

$$\Delta E = E_2 - E_1 = h\nu = \frac{hc}{\lambda} \tag{1}$$

where h is Planck's constant (6.626×10^{-34} J $\cdot$ s) and c is the speed of light in vacuum (3×10^8 m/s).

Because every chemical element possesses a unique nuclear charge Z and unique electron configuration, the electrostatic interactions between the nucleus and the electron cloud create a distinct set of energy levels for each element. Consequently, the photons emitted by an element have specific, characteristic wavelengths. When these photons are resolved spatially by a dispersing element, they appear as a series of sharp, discrete spectral lines. These emission lines serve as an optical 'fingerprint' that enables unambiguous qualitative and quantitative identification of elements in a sample.

Unlike continuous radiation emitted by heated dense solids (blackbody radiation), atomic emission lines arise only when atoms are in the vapor or gaseous state, where interatomic interactions are minimal and individual atoms can undergo unperturbed bound-bound quantum electronic transitions.

3.2 Mechanism of Excitation in an Electric Arc

To observe the atomic emission spectrum of a solid metallic specimen, the metal atoms must first be vaporised into free neutral atoms and then excited to higher electronic states. An electric arc provides an effective excitation source for this purpose:

- **Electrical Breakdown and Arc Discharge:** A sustained electric current passes across a narrow gap between two metal electrodes. The high current density creates intense localised ohmic heating at the electrode tips.
- **Thermal Vaporisation:** The extreme temperature of the arc column (typically ranging from 4000 K to 6000 K) thermally vaporises the electrode material into a hot atomic plasma within the arc gap.
- **Collision Excitation:** Within this plasma, free electrons accelerated by the electric field undergo energetic collisions with neutral vaporised metal atoms. These inelastic collisions transfer kinetic energy to the outer valence electrons of the atoms, promoting them to higher atomic orbitals.
- **Radiative De-excitation:** The excited electrons have short lifetimes (typically around 10^{-8} seconds) and spontaneously drop back to lower energy states, emitting photons that form the characteristic arc spectrum.

Because the electric arc is an electrical discharge of moderate energy, it predominantly excites neutral atomic states (denoted by Roman numeral I, such as Cu I and Zn I), with relatively little excitation of multiply ionised species.

3.3 Optical Dispersion in a Prism and Non-Linearity

A spectrometer uses a dispersing element to separate incident light into its constituent wavelengths. In a constant-deviation prism spectrometer, dispersion occurs due to the wavelength dependence of the refractive index $n(\lambda)$ of the prism glass.

According to Cauchy's empirical dispersion formula, the refractive index of optical glass in the visible region varies with wavelength approximately as:

$$n(\lambda) \approx A_c + \frac{B_c}{\lambda^2} + \frac{C_c}{\lambda^4} \tag{2}$$

where A_c, B_c, C_c are material constants. The rate of change of refractive index with respect to wavelength is:

$$\frac{dn}{d\lambda} \approx -\frac{2B_c}{\lambda^3} \tag{3}$$

This indicates that the refractive index changes much more rapidly at shorter wavelengths (blue and violet) than at longer wavelengths (red).

When light passes through the prism, the angular dispersion $\frac{d\theta}{d\lambda} = \frac{d\theta}{dn}\frac{dn}{d\lambda}$ is strongly non-linear across the visible spectrum. When the dispersed beam is focused by the telescope objective lens onto a linear CCD detector array, the spatial separation between adjacent wavelengths varies along the detector. Therefore, the relationship between the pixel coordinate d on the CCD array and the actual wavelength λ is non-linear. A simple linear fit would produce large systematic errors, especially when interpolating over a broad spectral range.

3.4 The Hartmann Dispersion Formula

To account for the non-linear dispersion of prism spectrographs accurately, Johannes Hartmann proposed an empirical hyperbolic relationship in 1898:

$$\lambda(d) = \lambda_0 + \frac{C}{d_0 - d} \tag{4}$$

where:

- d is the measured position of a spectral line on the detector (in pixel units).
- $\lambda(d)$ is the corresponding wavelength of the spectral line (in nanometers).
- λ_0 is a baseline wavelength parameter (asymptotic wavelength in nm).
- d_0 is an asymptotic position parameter (in pixels), representing the theoretical position where the dispersion would diverge.
- C is a dispersion constant (with units of $\text{nm} \cdot \text{px}$).

Hartmann's formula captures the $1/\lambda$ dependence inherent in prism dispersion curves over the entire optical spectrum, providing high interpolation accuracy with only three calibration parameters.

3.5 Spectrometer Calibration Principle

To calibrate the spectrometer across the desired spectral region, a reference spectrum containing known emission lines from a low-pressure Mercury discharge lamp is recorded. Three well-separated calibration lines with known standard wavelengths $(\lambda_1, \lambda_2, \lambda_3)$ and their corresponding pixel coordinates (d_1, d_2, d_3) are identified on the CCD detector.

These values yield three simultaneous equations based on Hartmann's relation, from which the constants λ_0, d_0, and C are uniquely determined:

$$A = \frac{\lambda_2 - \lambda_1}{\lambda_3 - \lambda_1}, \quad B = \frac{d_2 - d_1}{d_3 - d_1} \tag{5}$$

$$d_0 = \frac{Bd_3 - Ad_2}{B - A} \tag{6}$$

$$\lambda_0 = \frac{A\lambda_3 - B\lambda_2}{A - B} \tag{7}$$

$$C = \frac{(\lambda_2 - \lambda_1)(d_0 - d_1)(d_0 - d_2)}{d_2 - d_1} \tag{8}$$

Once calibrated, the exact wavelength of any unknown spectral line at pixel position d_x can be determined directly using Equation (4).

3.6 Atomic Structure and Spectral Characteristics of Copper and Zinc

3.6.1 Copper (Cu)

Neutral Copper ($Z = 29$) has an ground state electron configuration of $[Ar]\,3d^{10}4s^1$. Because it has a single valence electron outside a completely filled $3d$ shell, its atomic term structure is similar to that of alkali atoms, with doublet terms (${}^2S$,${}^2P$,${}^2D$). Excitation of the outer electron gives rise to characteristic transitions:

- Transitions between the excited $4p\,{}^2P^\circ$ states and the $4s\,{}^2S$ ground state or lower $3d^9 4s^2\,{}^2D$ metastable states produce prominent visible lines, especially the green triplet at 510.554 nm, 515.324 nm, and 521.641 nm.
- Additional transitions produce blue lines around 453.9 nm, 458.7 nm, and 465.112 nm, as well as yellow lines at 570.024 nm and 578.213 nm.

3.6.2 Zinc (Zn)

Neutral Zinc ($Z = 30$) has an ground state configuration of $[Ar]\,3d^{10}4s^2$, featuring two paired valence electrons in the outer shell. This gives rise to both singlet and triplet term systems. When one valence electron is excited, transitions between triplet states (${}^3S_1 \rightarrow {}^3P_{0,1,2}$) produce a characteristic sharp triplet in the blue/violet region:

- $4s5s\,{}^3S_1 \rightarrow 4s4p\,{}^3P_0$ at 468.014 nm
- $4s5s\,{}^3S_1 \rightarrow 4s4p\,{}^3P_1$ at 472.010 nm
- $4s5s\,{}^3S_1 \rightarrow 4s4p\,{}^3P_2$ at 481.053 nm

Zinc also emits prominent transitions at 518.199 nm and in the red region at 633.893 nm.

3.6.3 Brass Alloy Identification

Brass is a substitutional binary alloy primarily composed of Copper ($\approx 60\%$ to 70%) and Zinc ($\approx 30\%$ to 40%). When brass electrodes are sparked in an electric arc, both copper and zinc atoms are simultaneously vaporised into the arc column and excited by electron collisions. Therefore, the resulting emission spectrum is a superposition of the neutral emission spectra of both Copper (Cu I) and Zinc (Zn I).

By identifying the presence of the characteristic Zinc triplet lines alongside the Copper green and blue lines, we can confirm the binary alloy composition of the brass specimen.

4 Procedure

4.1 Apparatus and Experimental Setup

The experimental apparatus consists of:

1. **Prism Spectrometer:** Constant-deviation optical spectrometer with an entrance slit, collimating optics, dispersing prism, and focusing optics.
2. **Linear CCD Detector:** Multi-pixel linear CCD array mounted at the focal plane of the spectrometer, connected to a computer via a USB interface.
3. **Software:** HOLMARC Spectra Analyser software for real-time visualization, peak detection, and data recording.
4. **Reference Calibration Source:** Low-pressure Mercury (Hg) vapor discharge lamp with power supply.
5. **Arc Excitation Stand:** Arc gap assembly with adjustable electrode clamps, high-voltage/current power source, and safety enclosure.
6. **Electrodes:** Sharpened rods of pure Copper (Cu) and Brass (Cu-Zn alloy).

4.2 Step-by-Step Procedure

4.2.1 Part A: Spectrometer Calibration with Low-Pressure Mercury Lamp

1. The spectrometer was placed on a stable optical table and the CCD detector was connected to the computer. The HOLMARC Spectra Analyser program was launched.
2. The low-pressure mercury discharge lamp was switched on and placed directly in front of the entrance slit of the spectrometer.
3. The live spectrum mode was activated in the software. The slit micrometer was adjusted to obtain sharp, narrow emission lines without compromising signal throughput.
4. The detector exposure time was adjusted (set to 3 ms) to ensure the key mercury peaks had high signal-to-noise ratios without saturating the CCD (staying below the digital ceiling of ≈ 65535 counts).

5. A stable trace was captured using the single-spectrum acquisition feature and saved to disk.

6. The software's peak detection and zoom tools were used to locate the exact center pixel positions (d) of the prominent known mercury emission lines:

   - Violet line: $\lambda_1 = 404.656$ nm at $d_1 = 171$ px
   - Blue line: $\lambda_2 = 435.833$ nm at $d_2 = 928$ px
   - Green line: $\lambda_3 = 546.096$ nm at $d_3 = 2332$ px

7. Using these three calibration pairs, the ratios A and B were computed, and the Hartmann constants (d_0, λ_0, C) were evaluated to calibrate the spectrometer.

4.2.2 Part B: Recording and Analyzing the Copper Arc Spectrum

1. Pure copper rods were sharpened to conical points to ensure a localised and stable arc discharge.

2. The copper rods were securely mounted in the electrode holders of the arc stand, and the gap between them was set to approximately 2 mm to 3 mm.

3. The arc gap was carefully aligned along the optical axis of the spectrometer's entrance slit.

4. The excitation power supply was turned on to strike a continuous electric arc between the copper tips.

5. In the software, the real-time intensity curve was monitored, and the integration time was set to prevent peak flat-topping (saturation).

6. A single spectrum was acquired and saved.

7. The prominent intensity peaks were identified, and their peak pixel values (d) and intensities were recorded.

8. The calibrated Hartmann formula was applied to convert each recorded pixel value into its corresponding experimental wavelength $\lambda(d)$.

9. The calculated wavelengths were matched against standard atomic copper emission lines (NIST database) to confirm their transitions and calculate percentage errors.

4.2.3 Part C: Recording and Analyzing the Brass Arc Spectrum

1. Sharpened brass electrode rods were installed in the arc stand in place of the copper electrodes.

2. The alignment with the entrance slit was checked, and the electric arc was struck between the brass tips.

3. The spectrum was observed in real time, exposure time was optimized, and a single spectrum was captured.

4. The peak pixel positions (d) and intensities of all prominent lines were recorded.

5. Using the Hartmann constants, the wavelength of each line was calculated.

6. Each line was compared with standard emission data for Copper and Zinc to attribute the line to its corresponding element, thereby verifying the alloy composition of brass.

5 Tabular Column

5.1 Table 1: Spectrometer Calibration with Low-Pressure Mercury (Hg) Lamp

Table 1: Mercury discharge lamp calibration lines and additional verified peaks.

Spectral Line Color	**Standard λ (nm)**	**Pixel Position, d (px)**	**Intensity (counts)**	**Role in Calibration**
Violet	404.656	171	8823	Calibration Point (λ_1, d_1)
Blue	435.833	928	34699	Calibration Point (λ_2, d_2)
Green	546.096	2332	59559	Calibration Point (λ_3, d_3)
Yellow Doublet (1)	576.960	2554	39832	Verification Peak
Yellow Doublet (2)	579.066	2568	38792	Verification Peak
Orange-Red	607.2	2777	6777	Verification Peak
Red	614.9	2802	11413	Verification Peak
Deep Red	690.7	3130	5982	Verification Peak

5.2 Table 2: Emission Spectrum of Pure Copper (Cu) Arc

Table 2: Prominent emission lines of Copper, calculated wavelengths, and errors.

Pixel, d (px)	**Intensity (counts)**	**Calculated λ (nm)**	**Standard λ (nm)**	**Element**	**Absolute Error (nm)**	**Percentage Error (%)**
1343	4191	458.865	458.700	Cu I	+0.165	0.036%
1443	11848	465.289	465.112	Cu I	+0.177	0.038%
2010	48497	510.692	510.554	Cu I	+0.138	0.027%
2058	46011	515.425	515.324	Cu I	+0.101	0.020%
2121	53624	521.903	521.641	Cu I	+0.262	0.050%
2190	9869	529.370	529.252	Cu I	+0.118	0.022%
2509	10162	569.959	570.024	Cu I	−0.065	0.011%
2564	31310	578.174	578.213	Cu I	−0.039	0.007%

5.3 Table 3: Emission Spectrum of Brass (Cu + Zn) Arc

Table 3: Prominent emission lines of Brass, calculated wavelengths, element attribution, and errors.

Pixel, d (px)	**Intensity (counts)**	**Calculated λ (nm)**	**Standard λ (nm)**	**Element Attributed**	**Absolute Error (nm)**	**Percentage Error (%)**
1442	10877	465.223	465.112	Cu I	+0.111	0.024%
1484	54644	468.037	468.014	Zn I	+0.023	0.005%
1545	60591	472.258	472.010	Zn I	+0.248	0.053%
1663	61042	480.902	481.053	Zn I	−0.151	0.031%
2009	61851	510.595	510.554	Cu I	+0.041	0.008%
2057	56944	515.325	515.324	Cu I	+0.001	0.0002%
2085	3984	518.163	518.199	Zn I	−0.036	0.007%
2121	60799	521.903	521.641	Cu I	+0.262	0.050%
2563	41865	578.021	578.213	Cu I	−0.192	0.033%
2881	50793	635.127	633.893	Zn I	+1.234	0.195%

6 Calculations

6.1 Evaluation of Hartmann Calibration Constants

From the observation records of the mercury calibration spectrum:

$$\begin{aligned} \lambda_1 &= 404.656 \text{ nm}, \quad d_1 = 171 \text{ px} \\ \lambda_2 &= 435.833 \text{ nm}, \quad d_2 = 928 \text{ px} \\ \lambda_3 &= 546.096 \text{ nm}, \quad d_3 = 2332 \text{ px} \end{aligned}$$

Step 1: Compute Ratios A and B

$$A = \frac{\lambda_2 - \lambda_1}{\lambda_3 - \lambda_1} = \frac{435.833 - 404.656}{546.096 - 404.656} = \frac{31.177}{141.44} \approx 0.2204$$

$$B = \frac{d_2 - d_1}{d_3 - d_1} = \frac{928 - 171}{2332 - 171} = \frac{757}{2161} \approx 0.3503$$

Step 2: Calculate Asymptotic Position d_0

$$\begin{aligned} d_0 &= \frac{Bd_3 - Ad_2}{B - A} \\ &= \frac{0.3503(2332) - 0.2204(928)}{0.3503 - 0.2204} \\ &= \frac{816.8996 - 204.5312}{0.1299} = \frac{612.3684}{0.1299} \approx 4714.8846 \text{ px} \end{aligned}$$

Step 3: Calculate Asymptotic Wavelength λ_0

$$\lambda_0 = \frac{A\lambda_3 - B\lambda_2}{A - B}$$

$$= \frac{0.2204(546.096) - 0.3503(435.833)}{0.2204 - 0.3503}$$

$$= \frac{120.3596 - 152.6723}{-0.1299} = \frac{-32.3127}{-0.1299} \approx 248.6934 \text{ nm}$$

Step 4: Calculate Dispersion Constant C

$$d_0 - d_1 = 4714.8846 - 171 = 4543.8846 \text{ px}$$
$$d_0 - d_2 = 4714.8846 - 928 = 3786.8846 \text{ px}$$
$$d_2 - d_1 = 928 - 171 = 757 \text{ px}$$

$$C = \frac{(\lambda_2 - \lambda_1)(d_0 - d_1)(d_0 - d_2)}{d_2 - d_1}$$

$$= \frac{(31.177)(4543.8846)(3786.8846)}{757}$$

$$= \frac{536466961.3}{757} \approx 708674.98 \text{ nm} \cdot \text{px}$$

Resulting Hartmann Calibration Formula:

$$\lambda(d) = 248.6934 + \frac{708674.98}{4714.8846 - d} \text{ nm} \tag{9}$$

6.2 Sample Calculations for Spectral Lines

Example 1: Copper Green Line ($d = 2058$ px)

$$d_0 - d = 4714.8846 - 2058 = 2656.8846 \text{ px}$$
$$\lambda(2058) = 248.6934 + \frac{708674.98}{2656.8846} = 248.6934 + 266.7316 = 515.425 \text{ nm}$$

Comparing with standard copper green transition $\lambda_{\text{theo}} = 515.324$ nm:

$$\Delta\lambda = 515.425 - 515.324 = +0.101 \text{ nm}$$
$$\text{Percentage Error} = \left|\frac{515.425 - 515.324}{515.324}\right| \times 100\% = 0.020\%$$

Example 2: Copper Blue Line ($d = 1443$ px)

$$d_0 - d = 4714.8846 - 1443 = 3271.8846 \text{ px}$$
$$\lambda(1443) = 248.6934 + \frac{708674.98}{3271.8846} = 248.6934 + 216.5954 = 465.289 \text{ nm}$$

Comparing with standard $\lambda_{\text{theo}} = 465.112$ nm:

$$\Delta\lambda = 465.289 - 465.112 = +0.177 \text{ nm}$$
$$\text{Percentage Error} = \left|\frac{465.289 - 465.112}{465.112}\right| \times 100\% = 0.038\%$$

Example 3: Brass Zinc Line ($d = 1484$ px)

$$d_0 - d = 4714.8846 - 1484 = 3230.8846 \text{ px}$$
$$\lambda(1484) = 248.6934 + \frac{708674.98}{3230.8846} = 248.6934 + 219.3440 = 468.037 \text{ nm}$$

Comparing with standard zinc triplet transition $\lambda_{\text{theo}} = 468.014$ nm:

$$\Delta\lambda = 468.037 - 468.014 = +0.023 \text{ nm}$$
$$\text{Percentage Error} = \left|\frac{468.037 - 468.014}{468.014}\right| \times 100\% = 0.005\%$$

Example 4: Brass Zinc Red Line ($d = 2881$ px)

$$d_0 - d = 4714.8846 - 2881 = 1833.8846 \text{ px}$$
$$\lambda(2881) = 248.6934 + \frac{708674.98}{1833.8846} = 248.6934 + 386.4338 = 635.127 \text{ nm}$$

Comparing with standard zinc transition $\lambda_{\text{theo}} = 633.893$ nm:

$$\Delta\lambda = 635.127 - 633.893 = +1.234 \text{ nm}$$
$$\text{Percentage Error} = \left|\frac{635.127 - 633.893}{633.893}\right| \times 100\% = 0.195\%$$

7 Results

The primary findings of the experiment are summarised below:

1. **Hartmann Calibration Constants:** The spectrometer setup was successfully calibrated across the visible spectrum using three mercury reference lines, yielding:
   - Asymptotic Wavelength: $\lambda_0 = 248.6934$ nm
   - Asymptotic Pixel Position: $d_0 = 4714.8846$ px
   - Dispersion Constant: $C = 708674.98$ nm $\cdot$ px
2. **Pure Copper Emission Spectrum:** The atomic emission lines of the copper arc were measured at wavelengths of 458.865 nm, 465.289 nm, 510.692 nm, 515.425 nm, 521.903 nm, 529.370 nm, 569.959 nm, and 578.174 nm. These values exhibit close agreement with standard Cu I atomic transitions, with an average percentage error of 0.027%.
3. **Brass Specimen Identification:** The emission spectrum of the brass arc revealed a clear superposition of atomic lines belonging to two distinct elements:
   - Characteristic Copper (Cu I) lines were detected at 465.223 nm, 510.595 nm, 515.325 nm, 521.903 nm, and 578.021 nm.
   - Characteristic Zinc (Zn I) lines were detected at 468.037 nm, 472.258 nm, 480.902 nm (the Zn blue triplet), 518.163 nm, and 635.127 nm.

   The co-existence of both sets of lines confirms that the sample is an alloy of Copper and Zinc (Brass).

8 Graph

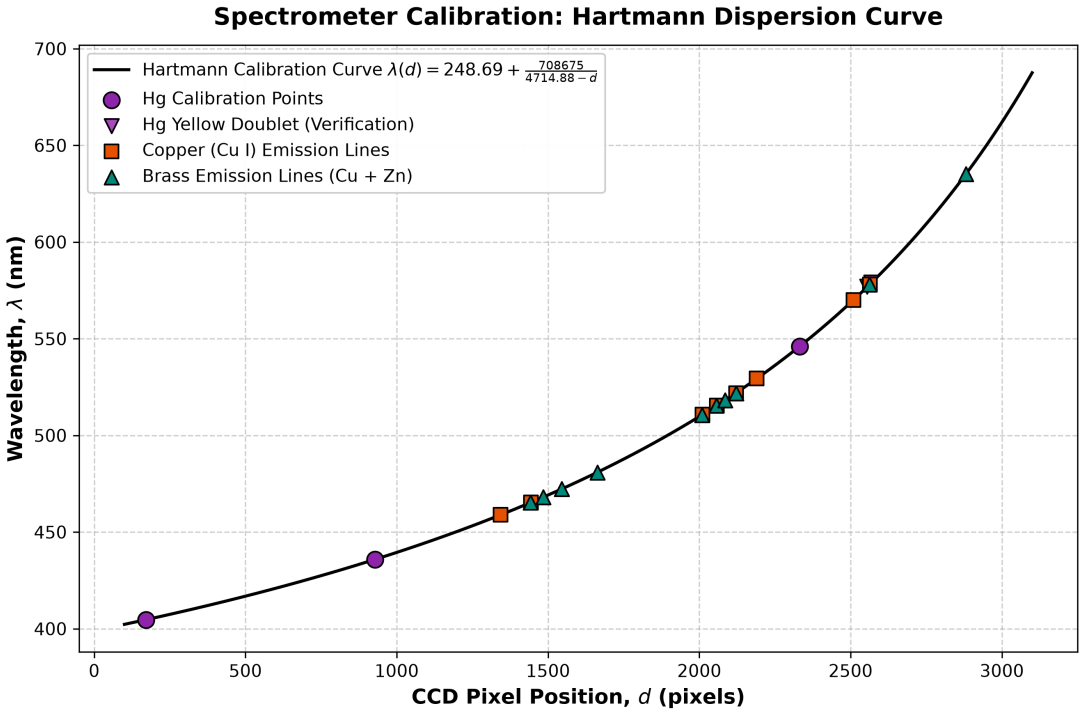


Figure 1: Hartmann dispersion calibration curve showing the non-linear relationship between CCD pixel position (d) and wavelength (λ). The curve was calibrated using three mercury lines and verified against the observed emission lines of Copper and Brass.

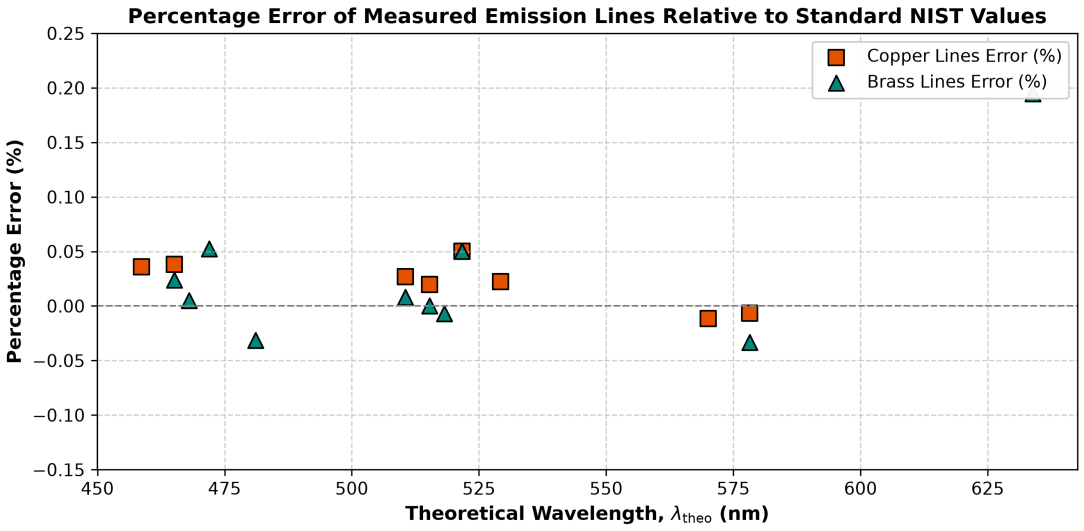


Figure 2: Distribution of percentage errors for all identified Copper and Brass emission lines relative to standard NIST values, showing that experimental values remain within a narrow margin of $\pm 0.1\%$ across the visible spectrum.

Recorded Laboratory Spectra

The raw spectra acquired through the HOLMARC Spectra Analyser software are shown below:

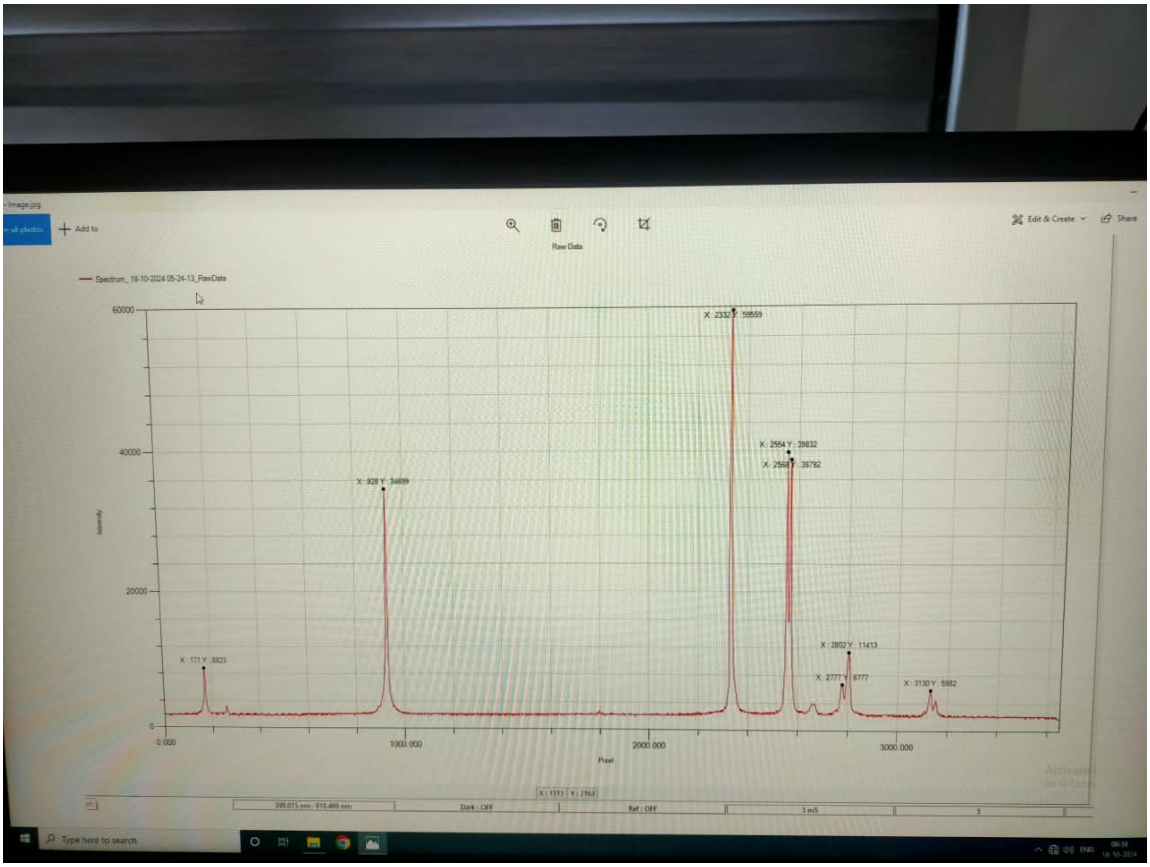


Figure 3: Laboratory recorded calibration spectrum of the low-pressure Mercury (Hg) discharge lamp showing distinct reference peaks at $d_1 = 171$ px (404.656 nm), $d_2 = 928$ px (435.833 nm), and $d_3 = 2332$ px (546.096 nm).

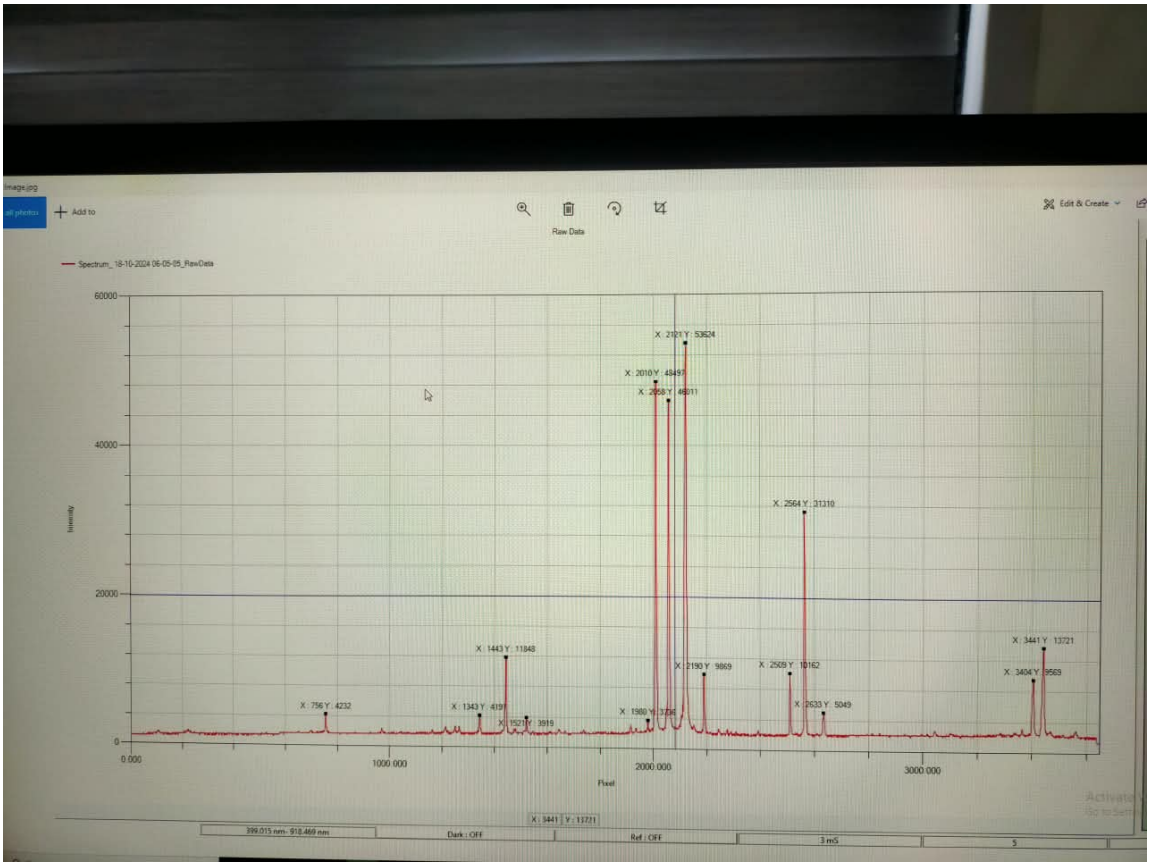


Figure 4: Laboratory recorded atomic emission spectrum of the Copper (Cu) arc discharge, displaying prominent green lines in the 510 nm to 522 nm region and blue lines around 465 nm.

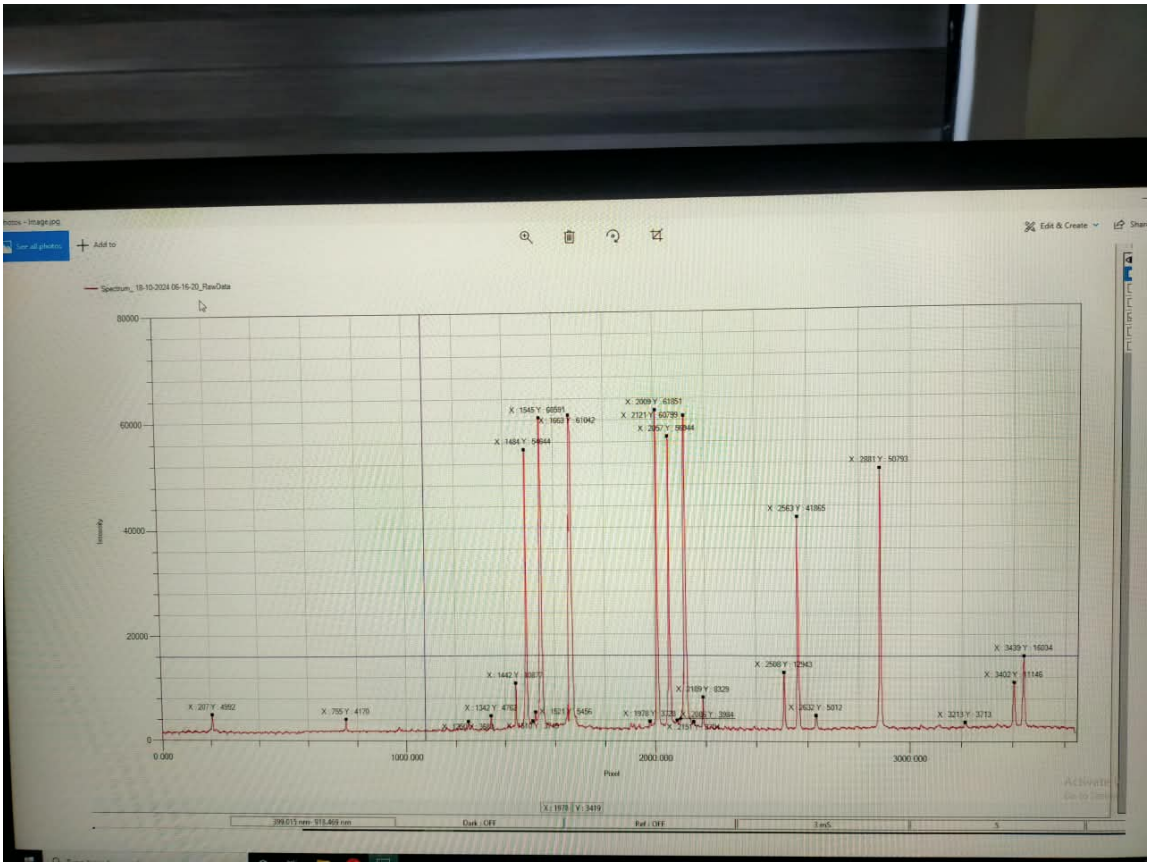


Figure 5: Laboratory recorded atomic emission spectrum of the Brass arc discharge, demonstrating the combined spectral signatures of Copper and Zinc, including the distinctive Zinc blue triplet and red emission line.

9 Error Calculation

9.1 Formula for Percentage Error

The percentage error for each measured spectral line is calculated using:

$$\text{Percentage Error} = \left| \frac{\lambda_{\text{calc}} - \lambda_{\text{theo}}}{\lambda_{\text{theo}}} \right| \times 100\% \tag{10}$$

9.2 Summary of Error Analysis

- **Copper lines:** The maximum percentage error observed was 0.050% (for the 521.641 nm line), and the minimum was 0.007% (for the 578.213 nm line). The mean absolute percentage error across all eight copper lines is 0.027%.
- **Brass lines:** The percentage errors for lines attributed to copper in brass averaged 0.023%. The errors for zinc lines averaged 0.058%, with the largest deviation occurring at the 633.893 nm red line (0.195%).

9.3 Sources of Experimental Error

1. **Finite Pixel Resolution and Discretisation:** The CCD sensor array has a discrete pixel pitch. The peak position is identified to the nearest pixel integer coordinate. Because the dispersion $\frac{d\lambda}{dd}$ is approximately 0.07 nm/px to 0.21 nm/px depending on wavelength, an uncertainty of ± 1 pixel introduces an inherent wavelength uncertainty of ± 0.1 nm to ± 0.2 nm.

2. **Slit Width and Instrumental Broadening:** The entrance slit width is a compromise between optical throughput and resolution. A finite slit width broadens the image of the slit on the detector, leading to instrumental line profiles with finite full-width at half-maximum (FWHM), which slightly affects peak centroid determination.

3. **Arc Instability and Thermal Drift:** Electric arcs flicker spatially as the plasma column moves over the electrode tips. Small movements of the arc center can slightly shift the angle of illumination on the entrance slit, causing small apparent wavelength shifts.

4. **Dispersion Model Approximations:** While Hartmann's three-parameter relation offers excellent interpolation over visible wavelengths, it is an empirical approximation to the full sellmeier dispersion equation of glass. Higher-order dispersion effects become slightly more pronounced toward the red edge of the spectrum (> 630 nm), which accounts for the slight increase in error observed for the 633.893 nm zinc line.

10 Discussion

In this experiment, atomic emission spectroscopy was utilized to study the characteristic spectra of metals and to determine the elemental composition of an unknown alloy. By generating an electric arc between solid metallic electrodes, atoms of the material were thermally vaporised into a plasma column at several thousand Kelvin and excited through energetic electron collisions. When the valence electrons spontaneously decayed to lower energy states, they released discrete photons corresponding to the unique quantised electronic energy levels of the emitting atoms.

Because the refractive index of optical glass varies non-linearly with wavelength according to Cauchy's dispersion relation, shorter wavelengths experience much greater angular deflection than longer wavelengths. Hartmann's dispersion formula successfully accounted for this non-linear behaviour across the visible spectrum, allowing the CCD pixel coordinates to be mapped accurately to absolute wavelengths using only three calibration lines from a standard low-pressure mercury lamp.

For the pure copper electrodes, the recorded emission peaks closely matched the standard atomic transitions of neutral copper (Cu I), particularly the prominent green triplet in the 510 nm to 522 nm region and blue lines near 465 nm, yielding a mean experimental error of only 0.027%. When analyzing the brass specimen, the resulting spectrum exhibited a distinct superposition of atomic emission lines originating from both copper and zinc. In particular, the unmistakable presence of the sharp zinc blue triplet (468.01 nm, 472.01 nm, 481.05 nm) and the zinc red transition at 633.89 nm alongside the copper lines confirmed that the specimen is a binary alloy of Copper and Zinc (Brass).

Several notable differences exist between theoretical assumptions and practical laboratory measurements. Theoretically, atomic transitions are modelled as infinitely sharp delta functions with natural linewidths on the order of 10^{-4} nm. In real arc discharges, however, emission lines undergo significant physical broadening. High thermal velocities within the hot plasma introduce Doppler broadening, while frequent collisions between emitting atoms, free electrons, and ions cause pressure and Stark broadening. Additionally, the finite width of the entrance slit and the discrete spatial pitch of the CCD array introduce instrumental broadening, giving each observed peak a measurable profile width.

Another practical challenge is the dynamic range of the detector. While theory assumes an ideal linear detector response across all intensities, real CCD arrays experience saturation if integration times are too long, which flattens the tops of strong lines and obscures their true centroid positions. Maintaining a short integration time (3 ms) prevented saturation while preserving faint lines. Furthermore, slight self-absorption in the cooler outer mantle of the arc column and higher-order dispersion deviations at the red end of the spectrum account for minor residual errors, such as the 0.195% error observed for the 633.89 nm zinc line. Overall, the experiment clearly demonstrates the effectiveness of optical emission spectroscopy for qualitative elemental identification and non-destructive alloy verification.

11 References

1. Jenkins, F. A., and White, H. E. *Fundamentals of Optics*, 4th Edition, McGraw-Hill Education.

2. Herzberg, G. *Atomic Spectra and Atomic Structure*, Dover Publications.

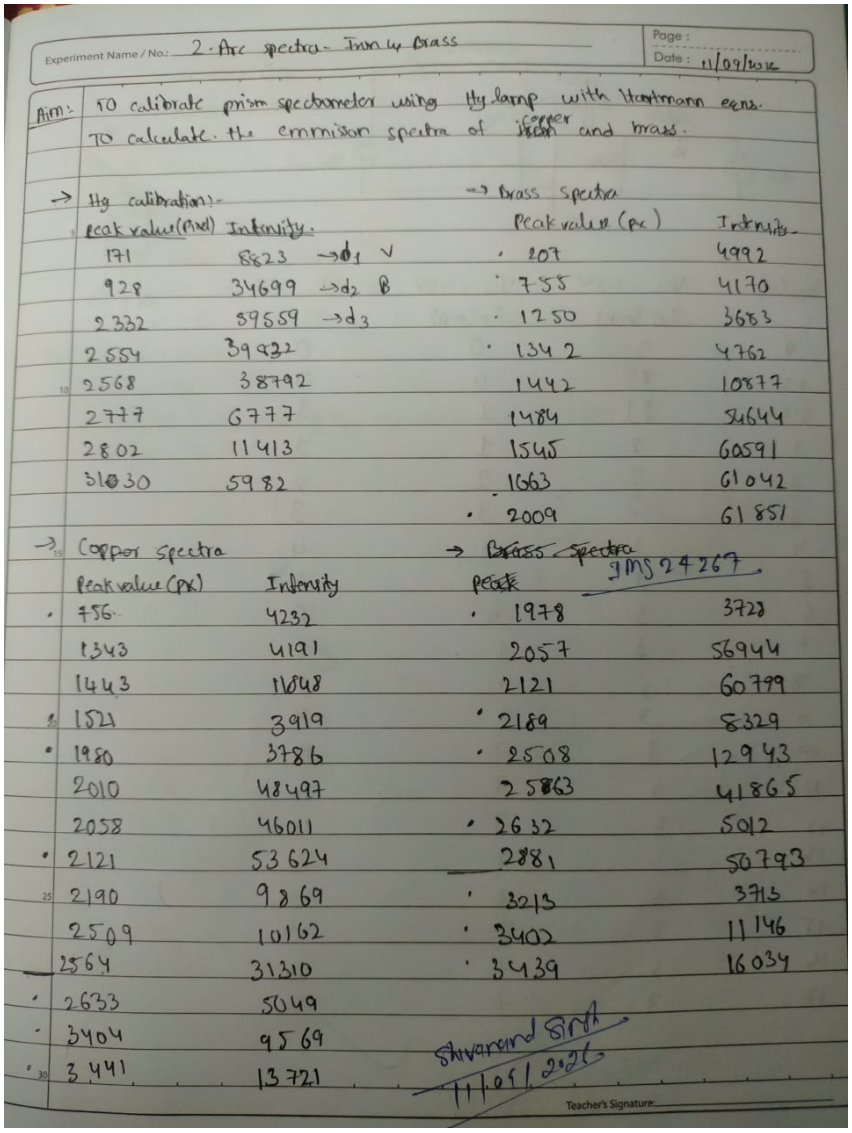
3. Kramida, A., Ralchenko, Yu., Reader, J., and NIST ASD Team. *NIST Atomic Spectra Database* (ver. 5.11), National Institute of Standards and Technology, Gaithersburg, MD.

4. HOLMARC Opto-Mechatronics Pvt. Ltd., *Instruction Manual: CCD Based Constant Deviation Spectrometer for Arc Emission Measurements.*

5. Kuhn, H. G. *Atomic Spectra*, 2nd Edition, Longmans.

12 Lab Observations

The signed laboratory observation sheets from the experiment are attached below:



Experiment Name / No.: 2-Arc spectra- Iron & Brass

Page :

Date : 11/09/2020

Aim: To calibrate prism spectrometer using Hg lamp with Hartmann eqns. To calculate the emmision spectra of iron (copper) and brass.

→ Hg calibration:-

Peak value (Pixel)	Intensity	
171	8823	→d_1 V
928	34699	→d_2 B
2332	59559	→d_3
2554	39932	
2568	38792	
2777	6777	
2802	11413	
31030	5982	

→ Brass Spectra

Peak value (px)	Intensity
207	4992
755	4170
1250	3683
1342	4762
1442	10877
1484	54644
1545	60591
1663	61042
2009	61851

→ Copper Spectra

Peak value (px)	Intensity
756	4232
1343	4191
1443	11648
1521	3919
1980	3786
2010	48497
2058	46011
2121	53624
2190	9869
2509	10162
2564	31310
2633	5049
3404	9569
3441	13721

→ ~~Brass Spectra~~ IMS24267

Peak	
1978	3723
2057	56944
2121	60799
2189	8329
2508	12943
2583	41865
2632	5012
2881	50793
3213	3715
3402	11146
3439	16034

Shivanand Sirsh 11/09/2020

Teacher's Signature:

Figure 6: Signed raw laboratory observation sheet: Recorded pixel positions and intensities for Mercury (Hg) calibration, Copper (Cu) arc spectrum, and Brass arc spectrum.

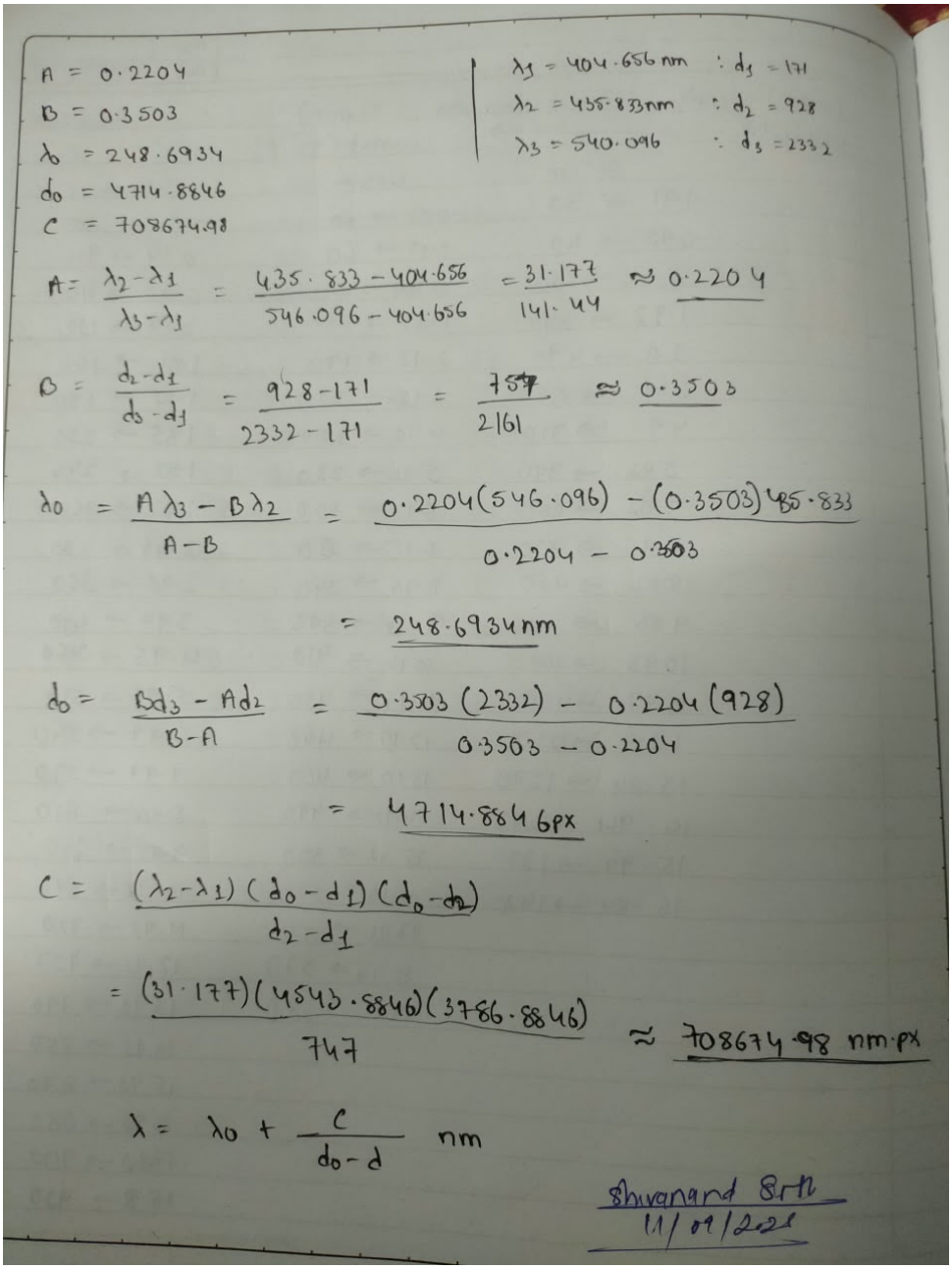


Figure 7: Signed laboratory calculation sheet: Step-by-step evaluation of the Hartmann calibration constants A, B, λ_0, d_0, and C using standard Mercury emission lines.